ModelIt!

Teacher's Guide

Climate-Ecosystem Coupled Models

Modeling Feedback Loops Between Climate Change, Ecosystems, and Human Society

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-ESS3-5, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-5, HS-LS2-6

Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change; evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Analyzing and Interpreting Data** — Students develop a coupled climate-ecosystem model with multiple interacting feedback loops, analyzing data from simulations to identify tipping points and project cascading consequences across natural and human systems.
Disciplinary Core Idea	**ESS3.D Global Climate Change / LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Human activities affect global climate through greenhouse gas emissions; ecosystems respond to climate disruption through species migration, productivity changes, and potential collapse when adaptation limits are exceeded.
Crosscutting Concept	**Stability and Change / Cause and Effect** — Students analyze how positive and negative feedback loops determine whether the climate-ecosystem system maintains stability or undergoes irreversible transitions, tracing causal chains across atmospheric, oceanic, terrestrial, and societal systems.

CAST Testing Alignment

MULTIPLE CHOICE:

Global Temperature increases by 3 degrees C above pre-industrial levels. Based on your model, which of the following feedback loops is most likely to become self-sustaining at this temperature, causing continued warming regardless of emission reductions?

CONSTRUCTED RESPONSE:

Using your coupled climate-ecosystem model, explain why reducing Atmospheric CO2 emissions to zero today would NOT immediately stop climate change. Reference at least four model components and describe the lag effects and committed changes already locked into the system.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to coupled climate-ecosystem modeling — the integration of atmospheric physics, ocean chemistry, terrestrial ecology, and human systems into a single interconnected model. Biotech skill focus: Multi-system feedback analysis and tipping point dynamics. Understanding the Earth as a coupled system is essential for making evidence-based decisions about climate mitigation and adaptation.

The Carbon Cycle and Climate System

The Earth's climate is regulated by the balance between incoming solar radiation and outgoing infrared radiation. Greenhouse gases — primarily CO2, methane (CH4), nitrous oxide (N2O), and water vapor — trap outgoing radiation, warming the surface. The carbon cycle moves carbon between four major reservoirs: the atmosphere (approximately 880 GtC), the ocean (approximately 38,000 GtC), terrestrial biomass and soil (approximately 4,000 GtC), and fossil fuels (approximately 10,000 GtC). Human activity has transferred approximately 440 GtC from the fossil fuel reservoir to the atmosphere since 1750, increasing atmospheric CO2 from 280 to 425 ppm. The ocean and terrestrial biosphere have absorbed approximately half of these emissions — but their absorption capacity is declining as the system warms.

Positive Feedback Loops and Tipping Points

Several positive feedback loops in the climate system can amplify warming beyond what greenhouse gas emissions alone would produce. The ice-albedo feedback: warming melts reflective ice, exposing dark ocean or land that absorbs more solar radiation, causing more warming. The permafrost carbon feedback: warming thaws Arctic permafrost containing 1,500 GtC of frozen organic matter, releasing CO2 and methane as it decomposes, causing more warming. The water vapor feedback: warming evaporates more water, and water vapor is a greenhouse gas, causing more warming (this is the single largest amplifying feedback). The forest die-off feedback: extreme warming causes drought, heat stress, and wildfire that kill forests, releasing stored carbon and eliminating future sequestration capacity. Each feedback has a potential tipping point — a temperature threshold beyond which it becomes self-sustaining regardless of future emission reductions.

Ecosystem Responses to Climate Change

Ecosystems respond to climate change through three primary mechanisms. Migration: species shift their ranges poleward (average 17 km per decade for terrestrial species) or to higher elevations (11 m per decade) to track suitable climate conditions. Adaptation: populations with genetic variation may evolve tolerance to new conditions over generations — but the rate of current climate change far exceeds the rate at which most species can adapt genetically. Collapse: when species cannot migrate fast enough or adapt, populations decline and ecosystems reorganize — coral reef bleaching (affecting 75% of global reefs), boreal forest die-off, and savanna desertification are already underway. Agricultural systems face the same pressures: moderate warming (1-2 degrees C) may increase yields for some temperate crops through longer growing seasons and CO2 fertilization, but warming beyond 2-3 degrees C reduces yields for most staple crops (wheat: -6% per degree C, corn: -7.4% per degree C globally).

Climate Justice and Adaptation Inequality

Climate change impacts are profoundly unequal. The nations that have contributed the most cumulative CO2 emissions (US, EU, China) are generally wealthier and better equipped to adapt — with resources for sea walls, disaster response, crop insurance, and technological innovation. The nations that have contributed the least (sub-Saharan Africa, South Pacific islands, Southeast Asia) face the most severe impacts — sea level rise threatening low-lying nations, extreme heat reducing agricultural productivity, and intensified tropical storms — with the fewest resources to adapt. This asymmetry between responsibility and vulnerability is the central challenge of climate justice. The model's Societal Adaptation Capacity component makes this inequality visible: a 0.5-meter sea level rise is manageable for the Netherlands (which has centuries of flood engineering

expertise and trillions of dollars of GDP) but existential for Tuvalu (population 12,000, maximum elevation 4.6 meters, GDP $63 million).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten components spanning atmospheric drivers (Atmospheric CO2), climate responses (Global Temperature, Ocean Absorption, Permafrost Methane Release), ecosystem responses (Forest Carbon Sequestration, Species Migration Rate), human system impacts (Crop Yield Change, Sea Level Rise, Extreme Weather Frequency), and societal capacity (Societal Adaptation Capacity).

E **Establish Relationships**

Students determine that Atmospheric CO2 and Societal Adaptation Capacity are the two primary external inputs — one representing the driver of climate change (controllable through emission policy) and the other representing humanity's capacity to respond (controllable through investment, policy, and technology). All other eight components emerge as interconnected responses with critical feedback loops.

V **Visualize & Model**

Students build a coupled climate-ecosystem model with explicit positive and negative feedback loops, visualizing how warming cascades through atmospheric, oceanic, terrestrial, and societal systems with nonlinear dynamics and potential tipping points.

E **Evaluate Outcomes**

Students run Paris-compliant, business-as-usual, and tipping-point-analysis scenarios to discover the temperature thresholds where positive feedbacks become self-sustaining and cascading consequences become unavoidable.

R **Revise & Extend**

Students add Albedo Feedback, Ocean Acidification, or Climate Migration Pressure to explore additional feedback loops and cascading consequences that increase model realism.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn‘t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Climate change doesn't just warm the planet. It triggers cascading feedback loops that can accelerate warming beyond our control. When permafrost melts and releases methane, which causes more warming, which melts more permafrost — that's a feedback loop. And some of them have tipping points."*

Watch For

Display before/after images of Arctic ice, coral reefs, and forests. Let the scale of change register before proceeding.

Slide 2: Learning Objectives

3 min

Say: *"Today you're Earth system scientists. You're going to model the most complex system we've covered — the entire planet. Ten components, multiple feedback loops, and tipping points that could change everything."*

Watch For

Pre-teach positive feedback loop and tipping point. Use the analogy: a positive feedback is like a snowball rolling downhill, getting bigger and faster; a tipping point is the edge of the cliff.

Slide 3: Big Question

3 min

Say: *"Here's the terrifying question: Is there a point where climate change becomes self-sustaining — where even if we stopped all emissions tomorrow, the feedbacks would keep warming the planet? That's the tipping point question. Let's find out."*

Watch For

Think-pair-share: Students brainstorm what 'self-sustaining feedback' means in the climate system. What processes could cause warming to continue even after emissions stop? Collect and categorize ideas.

Slide 4: LEVER Framework

2 min

Say: *"We're coupling multiple systems — atmosphere, ocean, land, ecosystems, agriculture, and human society — into a single model. The output of each system becomes the input of another. This is what real climate models do."*

Watch For

Preview LEVER steps. Emphasize that this model has multiple feedback LOOPS, not just linear chains — outputs circle back to modify inputs.

Slide 5: Activity 1: Components

8 min

Say: *"Two of these components are under human control. Eight are responses. But some of those responses loop BACK to amplify or dampen the original driver. Find the feedback loops."*

Watch For

Guide sorting of external (Atmospheric CO_2, Societal Adaptation Capacity) versus internal components. Then identify which internal components feed BACK to influence others — mapping the positive and negative feedback loops.

Slide 6: Activity 2: Connections

8 min

Say: *"The permafrost-methane-warming loop is the scariest positive feedback in the entire climate system. 1,500 gigatons of frozen carbon — twice what's currently in the atmosphere — waiting to be released if we warm the Arctic enough. Draw this loop."*

Watch For

Students trace each feedback loop: CO_2 to temperature to permafrost to more CO_2 (positive), CO_2 to temperature to forest growth to less CO_2 (negative, until tipping point). Identify which feedbacks are winning.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run the three scenarios. In Paris compliance, can you keep the feedbacks manageable? In business-as-usual, what breaks first? And in the tipping point analysis — find the temperature where the permafrost feedback becomes unstoppable."*

Watch For

Students predict, then simulate. The tipping point analysis is the key discovery — finding the temperature threshold where the system crosses from reversible to irreversible. Discuss what this means for climate policy urgency.

Slide 8: Discoveries

5 min

Say: *"You just modeled the most important system on Earth. The feedback loops are real. The tipping points are real. And the unequal distribution of impacts is real. The question is: what do we do about it?"*

Watch For

Connect model findings to real IPCC projections. Discuss climate justice — who is most affected, who has the least capacity to adapt, and who bears the most responsibility.

Slide 9: STEM Challenge

12 min

Say: *"A coastal city of 2 million people is facing a climate crisis. They need a 50-year plan. Mitigation, adaptation, resilience, equity — all of it. Use your model to design the plan that saves this city."*

Watch For

Groups design integrated climate strategies: emission reduction targets, infrastructure investments, nature-based solutions, vulnerable population protection, and monitoring systems. Present plans with model-derived evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atmospheric CO2, Societal Adaptation Capacity
- **INTERNAL:** Global Temperature, Ocean Absorption, Permafrost Methane Release, Forest Carbon Sequestration, Species Migration Rate, Crop Yield Change, Sea Level Rise, Extreme Weather Frequency

Relationships

- **Atmospheric CO2 to Global Temperature:** POSITIVE (+) — Higher atmospheric CO2 traps more outgoing infrared radiation through the greenhouse effect, increasing the planet's energy imbalance and raising Global Temperature — each doubling of CO2 produces approximately 3 degrees C of equilibrium warming (climate sensitivity).
- **Global Temperature to Ocean Absorption:** NEGATIVE (-) — Higher temperatures reduce the ocean's capacity to absorb CO2 because warm water holds less dissolved gas (Henry's Law) and because warming stratifies the ocean, reducing mixing between surface and deep layers that sequester carbon long-term.
- **Global Temperature to Permafrost Methane Release:** POSITIVE (+) — Higher temperatures thaw Arctic permafrost, exposing frozen organic matter to microbial decomposition that releases CO2 and methane — this is a POSITIVE FEEDBACK because the released greenhouse gases cause additional warming that thaws more permafrost.
- **Forest Carbon Sequestration to Atmospheric CO2:** NEGATIVE (-) — Greater forest growth and photosynthesis remove more CO2 from the atmosphere, reducing the atmospheric concentration — this is a NEGATIVE FEEDBACK that partially counteracts emissions, but it can reverse at high temperatures when forests die and release stored carbon.
- **Global Temperature to Extreme Weather Frequency:** POSITIVE (+) — Higher temperatures intensify the hydrological cycle (warmer air holds 7% more water vapor per degree C), increase ocean heat content that fuels tropical storms, and create more extreme temperature events — all increasing the frequency and intensity of weather extremes.
- **Extreme Weather Frequency to Societal Adaptation Capacity:** NEGATIVE (-) — More frequent extreme weather events damage infrastructure, displace populations, destroy crops, and consume disaster response resources, progressively degrading a society's capacity to adapt to ongoing and future climate impacts.

Key Discoveries

- The climate system contains multiple positive feedback loops that can amplify warming beyond what CO2 emissions alone would produce — permafrost methane release, reduced ocean absorption, and forest die-off can each become self-reinforcing above critical temperature thresholds
- Forest Carbon Sequestration is a negative feedback that partially counteracts warming, but it has a tipping point — above approximately 2.5-3.0 degrees C of warming, widespread forest die-off from heat stress, drought, and wildfire converts forests from carbon sinks to carbon sources, flipping the feedback from negative to positive
- Sea Level Rise and Extreme Weather Frequency are lagging indicators — they continue to accelerate for decades after emissions peak because of thermal inertia in the ocean and committed ice sheet melting, meaning current commitments have already locked in significant future impacts
- Societal Adaptation Capacity is the most unequally distributed component — nations that have contributed the least to atmospheric CO2 (sub-Saharan Africa, South Pacific island nations) face the most severe impacts with the fewest resources to adapt, creating a fundamental climate justice challenge

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Climate change is a gradual, linear process that we can slowly adjust to"	The model reveals that climate change contains TIPPING POINTS — critical thresholds beyond which the system shifts abruptly to a new state. These transitions are nonlinear, meaning the relationship between cause (CO2 emissions) and effect (climate impacts) is not proportional. A small additional temperature increase near a tipping point can trigger dramatically larger consequences than the same increase at lower temperatures. The permafrost methane feedback, ice sheet collapse, and forest die-off all have tipping point characteristics. This means that 'wait and see' approaches to climate policy risk crossing irreversible thresholds.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Technology will solve climate change so we don't need to reduce emissions"	While technology is essential — renewable energy, carbon capture, energy storage — no technology can reverse the effects of positive feedback loops that have already been triggered. The model shows that once Permafrost Methane Release becomes self-sustaining, no amount of direct air carbon capture (currently removing approximately 0.01 Mt CO2/year) can compensate for the approximately 1-2 Gt CO2-equivalent/year of additional emissions. Technology COMPLEMENTS emission reduction — it cannot REPLACE it. The math is simple: we emit approximately 40 Gt CO2/year; the most optimistic carbon capture projections remove approximately 10 Gt CO2/year by 2050; we need to reduce emissions AND deploy capture simultaneously.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate change only affects polar bears and distant future generations"	The model's downstream components (Crop Yield Change, Extreme Weather Frequency, Sea Level Rise) demonstrate that climate change is affecting human systems NOW — 2023 was the hottest year in recorded history, extreme weather events have increased 5x since the 1970s, and agricultural disruptions are already reducing food security in vulnerable regions. Sea Level Rise is displacing communities in the Marshall Islands, Bangladesh, and Louisiana TODAY. The model shows that the impacts accelerate over time, meaning every year of delay in emission reduction commits future generations to worse outcomes. This is not a distant, abstract problem — it is a present, concrete reality with measurable human costs.